Set 22: Time Series Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse monthly sales as a time series using a line chart, an advertising-budget scatter plot, an autocorrelation plot, and a Tableau dashboard with a trend line and product-wise pie chart.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Month	Sales (in $)
January	15,000
February	18,000
March	22,000
April	20,000
May	23,000

Assumed advertising budget (not given in the dataset):

Month	Advertising Budget (in $, assumed)
January	2,000
February	2,500
March	3,500
April	3,000
May	3,800

Assumed sales split by product (not given in the dataset):

Product	Total Sales (in $, assumed)	Share
Product A	44,100	45%
Product B	34,300	35%
Product C	19,600	20%

Assumption: Questions 2 and 3 refer to advertising budget and product-wise sales, which are not in the dataset. The two tables above were assumed so that the charts can be drawn. The conclusions from Q2 and Q3 apply to these assumed values only. Questions 1 and 4 use the given sales data.

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
```

```
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
sales <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
                 levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  # ASSUMED: advertising budget is not given in the dataset
  Ad_Budget = c(2000, 2500, 3500, 3000, 3800)
)
# ASSUMED: split of total sales (98,000) across three products
products <- data.frame(Product = c("Product A", "Product B", "Product C"),
                       Sales = c(44100, 34300, 19600))
write.csv(sales, "Monthly_Sales.csv", row.names = FALSE)  # for Tableau
write.csv(products, "Sales_by_Product.csv", row.names = FALSE)
```

Q1. Time Series Line Chart – Monthly Sales Trend

> *In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.*

R Program:

```
print(sales[, c("Month", "Sales")], row.names = FALSE)
cat("Change month to month:", diff(sales$Sales), "\n")
cat("Total sales:", sum(sales$Sales), "| Average:", mean(sales$Sales), "\n")
sales_ts <- ts(sales$Sales, start = c(2023, 1), frequency = 12)
print(sales_ts)
sales$Month_num <- 1:5
p_line <- ggplot(sales, aes(x = Month_num, y = Sales)) +
  geom_line(colour = "#0072B2", linewidth = 1.2) + geom_point(size = 4, colour = "#0072B2") +
  geom_text(aes(label = format(Sales, big.mark = ",")), vjust = -1.2, size = 3.8) +
  scale_x_continuous(breaks = 1:5, labels = levels(sales$Month)) +
  scale_y_continuous(limits = c(12000, 26000), labels = scales::label_comma()) +
  labs(title = "Monthly Sales Trend", x = "Month", y = "Sales (in $)") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_line)
ggsave("s22_q1.png", p_line, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
    Month Sales
  January 15000
 February 18000
    March 22000
    April 20000
      May 23000
Change month to month: 3000 4000 -2000 3000
Total sales: 98000 | Average: 19600
       Jan   Feb   Mar   Apr   May
2023 15000 18000 22000 20000 23000
```

Output:

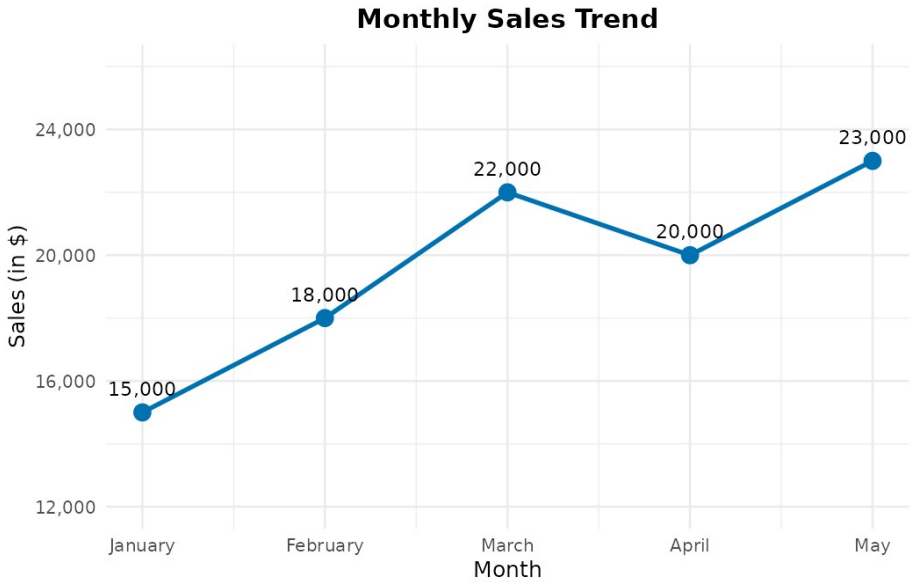


Figure 1: Monthly sales, January to May

Inference:

Sales show a clear **upward trend**, rising from $15,000 in January to $23,000 in May (+53.3%). The only fall is in April (−$2,000, −9.1%), after which sales recover to the highest value in May. Total sales are $98,000, an average of $19,600 per month.

Q2. Scatter Plot – Advertising Budget vs Monthly Sales

> *In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.*

R Program:

```
cat("Correlation (Ad_Budget vs Sales):", round(cor(sales$Ad_Budget, sales$Sales), 3), "\n")
fit <- lm(Sales ~ Ad_Budget, data = sales)
cat("Regression: Sales =", round(coef(fit)[1]), "+", round(coef(fit)[2], 2), "* Ad_Budget (R-
squared =", round(summary(fit)$r.squared, 3), ")\n")
p_sc <- ggplot(sales, aes(x = Ad_Budget, y = Sales)) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, colour = "grey50", linetype =
"dashed") +
  geom_point(size = 4.5, colour = "#0072B2") +
  geom_text(aes(label = Month), vjust = -1.4, size = 3.6) +
  scale_x_continuous(labels = scales::label_comma(), limits = c(1800, 4000)) +
  scale_y_continuous(labels = scales::label_comma(), limits = c(13000, 25000)) +
  labs(title = "Advertising Budget vs Monthly Sales", subtitle = "Budget values are assumed",
x = "Advertising Budget (in $)", y = "Sales (in $)") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_sc)
ggsave("s22_q2.png", p_sc, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Correlation (Ad_Budget vs Sales): 0.994
Regression: Sales = 6660 + 4.37 * Ad_Budget (R-squared = 0.989 )
```

Output:

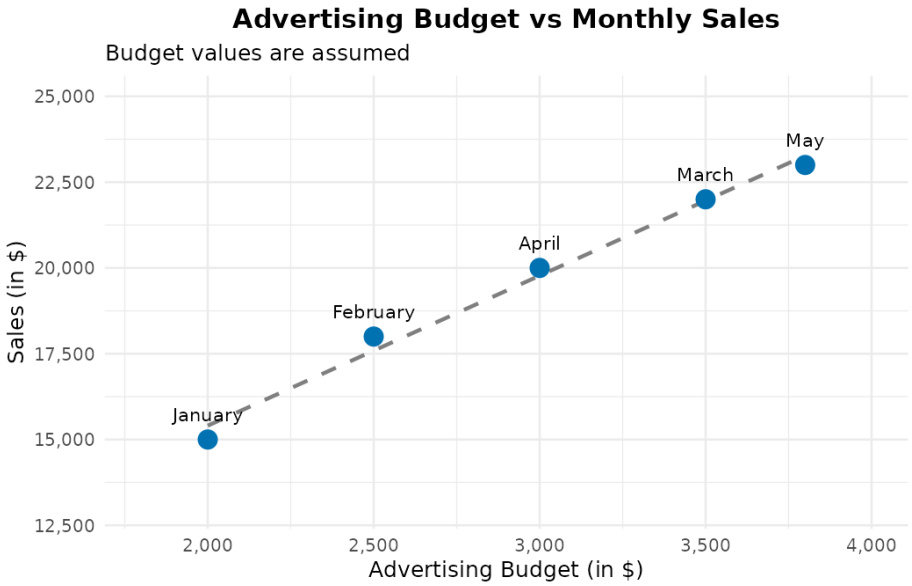


Figure 2: Advertising budget vs sales with a fitted line (budget values assumed)

Inference:

Using the assumed budgets, advertising and sales have a very strong positive relationship ($r = 0.994$, $R^2 = 0.989$). The fitted line suggests roughly **$4.37 of extra sales for each extra $1 of advertising**.

This is only a correlation from 5 assumed points. It does not prove advertising causes sales, since both may rise with seasonal demand, and the real budget data should be used before drawing conclusions.

Q3. Tableau Dashboard – Sales Trend Line and Product Pie Chart

> *In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.*

R Program:

```
products$Percent <- round(100 * products$Sales / sum(products$Sales), 1)
print(products, row.names = FALSE)
p_pie <- ggplot(products, aes(x = "", y = Sales, fill = Product)) +
  geom_col(width = 1, colour = "white") + coord_polar(theta = "y") +
  geom_text(aes(label = paste0(Product, "\n", Percent, "%")), position = position_stack(vjust
= 0.5),
            colour = "white", size = 3.6, fontface = "bold") +
  scale_fill_manual(values = c("Product A" = "#0072B2", "Product B" = "#E69F00", "Product C" =
"#009E73")) +
  labs(title = "Distribution of Sales across Products") +
  theme_void(base_size = 11) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
dash <- (p_line | p_pie) +
  plot_annotation(title = "Monthly Sales Dashboard", theme = theme(plot.title =
element_text(face = "bold", size = 15)))
ggsave("s22_dash.png", dash, width = 12, height = 4.8, dpi = 130, bg = "white")
ggsave("s22_pie.png", p_pie, width = 5.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
   Product Sales Percent
 Product A 44100      45
 Product B 34300      35
 Product C 19600      20
```

Output:

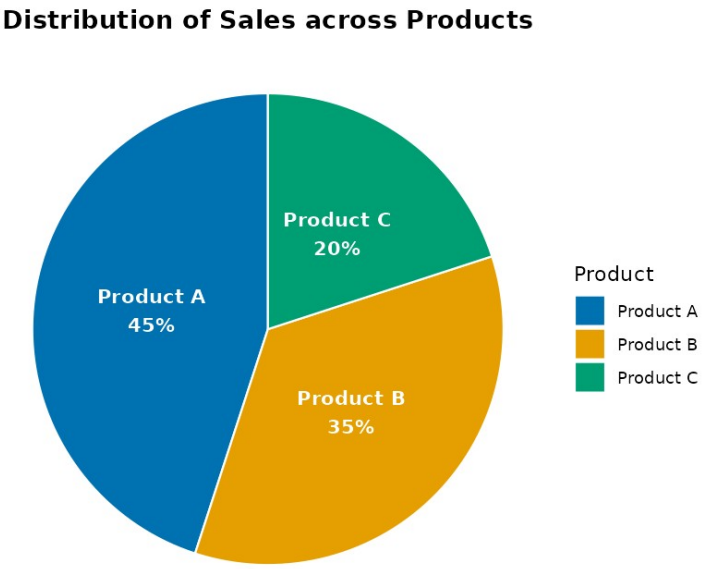
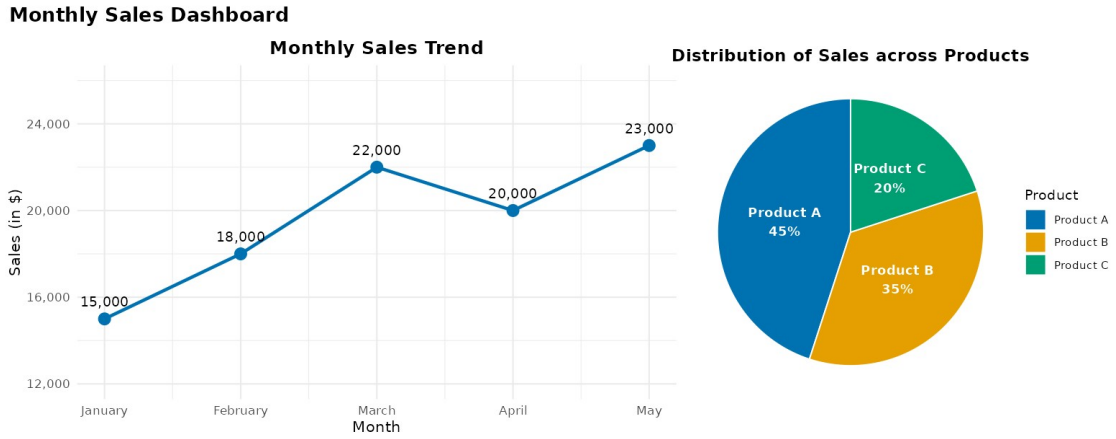


Figure 3: Distribution of sales across products (assumed split)

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → **Monthly_Sales.csv**. Then add **Sales_by_Product.csv** as a second data source (the sheets use separate sources).
2. **Sheet 1 – "Sales Trend":** drag Month to Columns (custom sort Jan to May) and SUM(Sales) to Rows. Set the Marks type to **Line** with markers and labels. Add **Analytics → Trend Line → Linear** to show the upward direction.
3. **Sheet 2 – "Sales by Product":** on the product data source set the Marks type to **Pie**. Drag Product to **Color** and SUM(Sales) to **Angle**. Show Product and **Percent of Total** as labels.
4. From the menu choose **Dashboard → New Dashboard** and set **Size → Automatic** (or Fixed 1200 × 800).
5. Drag the sheets onto the canvas: Sales Trend (left) and Sales by Product (right). Add a **Text** object at the top as the dashboard title.
6. On each chart, open the sheet's drop-down and select **Use as Filter**, so clicking a bar, point or slice filters the other views.
7. Go to **Dashboard → Actions → Add Action → Highlight** (Run on: Hover) so related marks highlight across all views.
8. Right-click each quick filter → **Apply to Worksheets → All Using This Data Source** (the relevant data source) so one filter controls every view.
9. Save as a packaged workbook (**File → Save As → .twbx**) or publish to Tableau Public.

Dashboard Layout:



Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

The line shows how sales grow over the five months, and the pie shows how the total is shared between products (A 45%, B 35%, C 20%, assumed). Clicking a product slice highlights its share and lets the viewer compare it with the monthly trend.

Q4. Autocorrelation Plot – Identifying Seasonality

> *In R, create an autocorrelation plot to identify seasonality in the time series data.*

R Program:

```
a <- acf(sales_ts, lag.max = 3, plot = FALSE)
print(a)
bound <- 1.96 / sqrt(length(sales_ts))
cat("95% significance bound: +/-", round(bound, 3), "\n")
acf_df <- data.frame(Lag = as.numeric(a$lag[-1]) * 12, ACF = as.numeric(a$acf[-1]))
acf_df$Lag <- 1:3
p_acf <- ggplot(acf_df, aes(x = Lag, y = ACF)) +
  geom_hline(yintercept = 0) +
  geom_hline(yintercept = c(-bound, bound), linetype = "dashed", colour = "#D55E00") +
  geom_segment(aes(xend = Lag, yend = 0), linewidth = 1.6, colour = "#0072B2") +
  geom_text(aes(label = round(ACF, 2)), vjust = -0.8, size = 4) +
  scale_x_continuous(breaks = 1:3) + scale_y_continuous(limits = c(-1, 1)) +
  labs(title = "Autocorrelation of Monthly Sales", subtitle = "Dashed lines = 95% significance
bounds", x = "Lag (months)", y = "ACF") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_acf)
ggsave("s22_q4.png", p_acf, width = 6.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```

Autocorrelations of series ‘sales_ts’, by lag

0.0000 0.0833 0.1667 0.2500
 1.000  0.142 -0.085 -0.177
95% significance bound: +/- 0.877
```

Output:

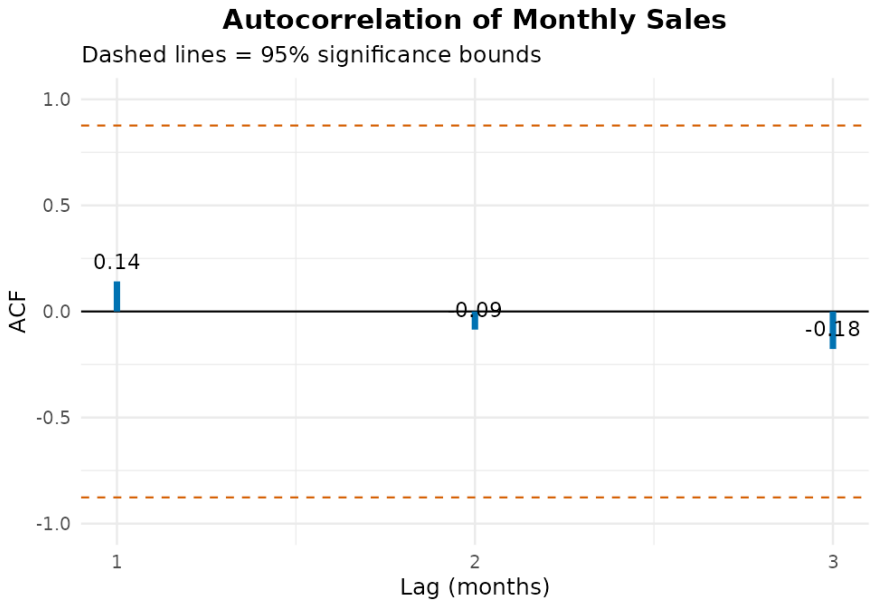


Figure 4: Autocorrelation function (ACF) of monthly sales

Inference:

The autocorrelations are **0.14 at lag 1, −0.09 at lag 2 and −0.18 at lag 3**. All lie well inside the 95% significance bounds (±0.877), so there is **no significant autocorrelation and no evidence of seasonality**. (R labels the lags in years, so 0.0833 equals one month.)

Five months of data cover less than one yearly cycle, so seasonality cannot be detected reliably. The rising trend dominates the series. At least 24 months (two full yearly cycles) are needed to see whether a spike appears at lag 12.

Result

Monthly sales rose from $15,000 to $23,000 over five months, with one dip in April. The assumed advertising budget is strongly correlated with sales (r = 0.994), and the assumed product split is A 45%, B 35% and C 20%. The autocorrelation plot shows no significant seasonality, but five months of data are too few to test it reliably.